
FAIRNESS, EFFICIENCY, AND STRATEGIC BEHAVIOUR IN SHARED GPU ALLOCATION

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ABSTRACT

Shared GPU clusters have to divide scarce compute among users whose job values are private and whose reports may be strategic. We study this problem in a repeated, unit-demand allocation model. The evaluation measures allocative efficiency, long-run fairness, and resistance to manipulation. A common simulator compares five mechanisms: random allocation, round-robin, greedy allocation by reported value, a history-penalised scoring rule, and a k -unit Vickrey auction. Reports are bounded. Payment and non-payment mechanisms use separate models, and all comparisons use paired valuation traces. The experiments vary scarcity, the fairness penalty, the fraction and strategy of misreporting users, valuation heterogeneity, temporal persistence, and population size. We will estimate the resulting frontier over welfare, fairness, and manipulation gain. The output will be a reproducible stress test of simple allocation rules, with evidence about when a modest fairness correction helps, fails, or introduces new strategic incentives.

Keywords GPU allocation · fair resource allocation · mechanism design · strategic behaviour · multi-agent systems

1 Introduction

Modern machine-learning workloads compete for expensive shared accelerators. The scheduling rule affects cluster efficiency and the time each user waits. Existing systems handle priorities, heterogeneous accelerators, placement, and long-running jobs [1, 2]. Fair access still matters. Repeatedly serving the jobs with the largest apparent benefit can starve other users. A rigid equal-time rule can also waste compute when job values differ substantially.

Private information creates another difficulty. If allocation depends on a self-reported priority, a user may inflate the report to receive service. The scheduler sees the claim, not the user’s true value. A rule that is efficient with truthful reports can therefore perform poorly when users act strategically. The analysis draws on online resource allocation, fair division, and mechanism design [3, 4, 5].

Several GPU schedulers provide useful points of comparison. Tiresias uses attained service to reduce job completion time. Gavel extends scheduling policies to heterogeneous accelerators. Themis combines finish-time fairness with a round-by-round partial-allocation auction designed to encourage truthful bids. Shockwave uses market-based planning for jobs that adapt over time [1, 2, 6, 7]. This project studies a narrower question. It uses a stylised unit-demand model to separate the effects of scarcity, allocation history, and bounded misreporting while holding information and random inputs constant. It does not claim a new production scheduler or a general impossibility theorem.

Research question.

How do simple repeated GPU-allocation mechanisms trade off true social welfare, long-run fairness, and empirical resistance to bounded strategic reporting as scarcity, user heterogeneity, and temporal demand patterns vary?

The experiments address three subsidiary questions:

1. Does a moderate history penalty reduce starvation without a large loss of true welfare?

2. Does increasing the fairness penalty change the benefit available from strategically timing or inflating reports?
3. Do conclusions based on equal allocation counts survive when users have unequal value distributions or persistent demand?

Research contribution. The project places allocation and payments in one formal model. It measures manipulation gain with paired counterfactual simulations and evaluates welfare, fairness, and manipulation together. Validation tests, confidence intervals, saved seeds, and common workloads make the experiments reproducible. The study is an analysis of mechanisms under controlled workloads, not a systems implementation.

2 Preliminaries

Agents, resources, and time. Let $[n] = \{1, \dots, n\}$ denote the unit-demand users. In each of T discrete rounds, they compete for $k < n$ identical GPUs. A user can receive at most one GPU in a round. Since all values are non-negative, the base model allocates all k GPUs. A seeded random generator breaks ties uniformly.

Valuations and reports. User i draws a private value $v_{i,t} \in [0, v_{\max}]$ from distribution $\mathcal{D}_i$ for receiving one GPU in round t . The user's report is $\hat{v}_{i,t} \in [0, v_{\max}]$. The common bound on values and reports rules out the degenerate case in which an arbitrarily large bid controls the allocation. Unless stated otherwise, $v_{\max} = 1$.

Allocations, payments, and histories. An allocation $\mathbf{x}_t \in \{0, 1\}^n$ satisfies $\sum_i x_{i,t} = k$. Its payment vector $\mathbf{p}_t \in \mathbb{R}_{\geq 0}^n$ has $p_{i,t} = 0$ whenever $x_{i,t} = 0$. For non-monetary mechanisms, $\mathbf{p}_t = \mathbf{0}$. The public history before round t is $h_t = (\mathbf{x}_1, \dots, \mathbf{x}_{t-1})$, and

$$a_i(t) = \sum_{\tau=1}^{t-1} x_{i,\tau}$$

is user i 's cumulative allocation. Let $q_i(t)$ be the number of consecutive rounds since user i last received a GPU.

Utility and welfare. Under quasi-linear utility, user i receives

$$u_{i,t} = v_{i,t}x_{i,t} - p_{i,t}, \quad U_i = \sum_{t=1}^T u_{i,t}.$$

The gross allocation benefit is $G_i = \sum_t v_{i,t}x_{i,t}$. Payments count as transfers when calculating social welfare:

$$W = \sum_{i=1}^n G_i = \sum_{t=1}^T \sum_{i=1}^n v_{i,t}x_{i,t}.$$

In round t , let S_t^* contain the k users with the highest true values. The truthful allocation oracle obtains

$$W^* = \sum_{t=1}^T \sum_{i \in S_t^*} v_{i,t}, \quad \text{WR} = W/W^*.$$

Allocation and benefit fairness. Let $A_i = a_i(T+1)$. Jain's allocation-fairness index is [8]

$$J_A = \frac{(\sum_i A_i)^2}{n \sum_i A_i^2}.$$

When the distributions $\mathcal{D}_i$ differ, equal access need not yield equal benefit. We also report Jain's index J_B on normalised gross benefits:

$$B_i = \frac{G_i}{T\mu_i}, \quad \mu_i = \mathbb{E}_{v \sim \mathcal{D}_i}[v], \quad J_B = \frac{(\sum_i B_i)^2}{n \sum_i B_i^2}.$$

Nash gross welfare is $\text{NSW} = \sum_i \log(G_i + \varepsilon)$ with $\varepsilon = 10^{-8}$.

Waiting and starvation. The maximum wait is $Q_{\max} = \max_{i,t} q_i(t)$. For $\Delta = 2\lceil n/k \rceil$, twice the service cycle of round-robin, the starvation rate is

$$\text{SR}_\Delta = \frac{1}{nT} \sum_{i=1}^n \sum_{t=1}^T \mathbb{I}\{q_i(t) > \Delta\}.$$

We report the 95th percentile of all waiting times with this threshold-based measure.

Paired strategic effects. Let ω contain a fixed valuation trace, tie-breaking seed, and set of strategic users. To evaluate manipulation gain, we change only the focal user’s strategy and hold ω and every other policy fixed:

$$M_i(\sigma_i; \omega) = U_i(\sigma_i, \sigma_{-i}; \omega) - U_i(\text{truthful}, \sigma_{-i}; \omega).$$

The reported statistics are the mean $\overline{M}$, the maximum $M_{\max}$, and the fraction of strategic users with $M_i > 0$. For a strategic profile, the welfare loss is

$$\text{PoS} = \frac{W_{\text{truthful}} - W_{\text{strategic}}}{W_{\text{truthful}}},$$

Both simulations use the same ω . For a fair mechanism F , the normalised welfare loss with truthful reports is

$$\text{PoF}(F) = \frac{W^* - W_F}{W^*}.$$

Incentive terminology. A one-round direct mechanism is dominant-strategy incentive compatible (DSIC) when truthful reporting weakly maximises each user’s utility for every profile of other reports [3]. Reports cannot affect a report-invariant rule, which makes such a rule weakly strategyproof. For history-dependent non-payment rules, the experiments measure vulnerability to specified attacks. Simulation cannot establish dynamic strategyproofness.

3 Problem formulation

Let $\rho \in [0, 1]$, and let $\mathcal{S} \subseteq [n]$, $|\mathcal{S}| = \lfloor \rho n \rfloor$, be the strategic set. A reporting policy may condition on the current value and public history:

$$\sigma_i : [0, v_{\max}] \times \mathcal{H}_t \longrightarrow [0, v_{\max}], \quad \hat{v}_{i,t} = \sigma_i(v_{i,t}, h_t).$$

A repeated mechanism is a sequence of mappings

$$\mathcal{M}_t : [0, v_{\max}]^n \times \mathcal{H}_t \longrightarrow \mathcal{X}_k \times \mathbb{R}_{\geq 0}^n, \quad \mathcal{X}_k = \left\{ \mathbf{x} \in \{0, 1\}^n : \sum_i x_i = k \right\},$$

which returns an allocation $\mathbf{x}_t$ and a payment vector $\mathbf{p}_t$. The same model therefore covers the non-payment rules and the auction.

For each workload condition, we estimate the achievable set

$$\mathcal{F} = \{(\text{WR}, J_A, J_B, \text{SR}_{\Delta}, \overline{M}) : \mathcal{M} \in \mathfrak{M}\},$$

where $\mathfrak{M}$ contains the five candidate mechanisms and the parameter settings described below. One setting Pareto dominates another when it has no lower welfare or fairness, no higher starvation or manipulation gain, and a strict improvement on at least one measure. Keeping the measures separate avoids an arbitrary scalar weighting of these normative objectives.

The base environment is a repeated allocation model, not a production scheduler. It assumes identical GPUs, one-round service, at most one GPU per user, and additive values. It omits job arrivals, gang scheduling, placement, preemption, and multi-GPU speedup. These restrictions allow controlled causal comparisons between mechanisms. A separate experiment introduces temporal persistence to test whether the conclusions depend on independent values.

4 Proposed solutions

4.1 Allocation mechanisms

Table 1: Properties evaluated or guaranteed by construction.

Mechanism	Reports	History	Payments	Formal property
M1 Random	No	No	No	Report-invariant
M2 Round-robin	No	Yes	No	Report-invariant
M3 Greedy	Yes	No	No	Truthful welfare oracle only
M4 Score	Yes	Yes	No	Empirically stress-tested
M5 Vickrey	Yes	No	Yes	Per-round DSIC

M1 Random. M1 chooses k users uniformly without replacement. It ignores reports and history. It is report-invariant and equalises expected allocation counts.

M2 Round-robin. M2 starts from a seeded queue and serves the next k users in each round. It is report-invariant and bounds waiting time, but ignores current job value.

M3 Greedy reported value. Allocate to the k largest reports:

$$\mathbf{x}_t \in \arg \max_{\mathbf{x} \in \mathcal{X}_k} \sum_{i=1}^n \hat{v}_{i,t} x_i.$$

When reports equal values, M3 attains the truthful welfare oracle. Increasing a report can only improve the user's one-round allocation probability.

M4 History-penalised score. For $\lambda \geq 0$, define

$$s_{i,t} = \frac{\hat{v}_{i,t}}{(1 + a_i(t))^\lambda}$$

and allocate to the k largest scores. M4 equals M3 at $\lambda = 0$. As λ grows, cumulative service has more influence relative to reported value. The rule approaches least-served-first, with reports used to break ties, but never becomes literal round-robin. Sweeping λ measures the fairness benefit and strategic cost of the correction.

M5 k -unit Vickrey auction. Allocate to the k highest bids and charge each winner the $(k + 1)$ st-highest bid:

$$p_{i,t} = x_{i,t} \hat{v}_{(k+1),t}.$$

With unit demand, quasi-linear utility, and no intertemporal budget, this is the VCG outcome for k identical units and is DSIC in each round [9, 3]. The model treats payments as utility-equivalent transfers. They are not reusable virtual credits. Real transfers are required for this truthfulness benchmark, which we use only as a mechanism-design baseline and do not propose as a deployment policy.

4.2 Strategic policies

Each strategic user follows one of four bounded policies:

1. **Truthful:** $\hat{v}_{i,t} = v_{i,t}$.
2. **Capped exaggeration:** $\hat{v}_{i,t} = \min\{c v_{i,t}, v_{\max}\}$ for $c \in \{1.25, 1.5, 2\}$.
3. **Maximum claim:** $\hat{v}_{i,t} = v_{\max}$, a worst-case priority-inflation baseline.
4. **Finite-horizon rollout attack:** for a diagnostic setting with $n = 10$, evaluate reports on $\mathcal{G} = \{0, 0.1, \dots, 1\}$ using 100 Monte Carlo rollouts over the next $H = 5$ rounds, and select the report with highest estimated cumulative utility under fixed opponent policies.

The rollout policy approximates a unilateral attack. It is neither an equilibrium computation nor a proof of manipulability. We report it separately because a finite search heuristic is not a true best response.

4.3 Experimental protocol

Unless a factor is being varied, the base configuration is

$$n = 50, \quad k = 10, \quad T = 1000, \quad \mathcal{D}_i = \text{Uniform}(0, 1), \quad \rho = 0.25.$$

Each condition uses 30 independent master seeds. For a given seed, every mechanism receives the same valuation tensor, strategic-user set, and pre-generated tie-breaking randomness. Reported results will include seed means and paired 95% bootstrap confidence intervals based on 10,000 resamples. We will save raw per-seed results, configuration files, and the master seed list. Experiment E3 uses a factorial design; all other experiments change one factor at a time.

E0 validation and sanity checks. Small discrete instances allow exhaustive checks of capacity and payment invariants, report-invariance of M1–M2, equivalence of M3 and M4 at $\lambda = 0$, the round-robin wait bound, welfare-oracle calculations, and one-round truthfulness of M5 for every report in a finite grid. These checks identify implementation errors before the main experiments run.

E1 resource scarcity. Vary $k/n \in \{0.1, 0.2, 0.4, 0.6, 0.8\}$ and compare WR, J_A , $Q_{\max}$, the 95th-percentile wait, and SR_{Δ} under truthful reports.

E2 fairness and strategic-vulnerability frontier. Sweep $\lambda \in \{0, 0.05, 0.1, 0.25, 0.5, 1, 2, 5\}$. Plot WR against J_A and colour each point by $\bar{M}$ under capped exaggeration. Report all non-dominated settings. No single metric will determine the selected value of λ .

E3 strategic population and attack type. Use a factorial design over $\rho \in \{0, 0.1, 0.25, 0.5, 1\}$ and the three scalable policies (truthful, capped exaggeration with $c = 2$, and maximum claim). Measure PoS, $\bar{M}$, $M_{\max}$, the fraction benefiting from manipulation, WR, and fairness. Apply the rollout attack to selected settings for M3–M5 where it is most likely to separate their behaviour.

E4 heterogeneous users. Compare the homogeneous base case with an equal-sized two-group population: $\mathcal{D}_L = \text{Beta}(2, 5)$ and $\mathcal{D}_H = \text{Beta}(5, 2)$. Both remain in $[0, 1]$ but have different means. Report J_A and J_B together to identify cases in which equal service and equal normalised benefit diverge.

E5 temporally persistent demand. Replace i.i.d. values with

$$v_{i,t} = \alpha v_{i,t-1} + (1 - \alpha)\epsilon_{i,t}, \quad \epsilon_{i,t} \sim \text{Uniform}(0, 1),$$

for $\alpha \in \{0, 0.5, 0.9\}$. The experiment checks whether the conclusions hold when high-value demand persists across rounds, a feature absent from the i.i.d. model.

E6 scalability. Vary $n \in \{10, 25, 50, 100, 250, 500\}$ with $k/n = 0.2$ and $T = 1000$. Report wall-clock time per simulated round and peak memory. This experiment excludes the rollout attack, which solves a separate and more expensive diagnostic problem.

4.4 Testable hypotheses

1. Moderate λ will reduce tail waiting and starvation before it causes a comparable loss in WR.
2. Increasing λ will improve allocation equality but may increase the value of strategically timing reports because current service changes future scores.
3. Round-robin will score well on J_A but worse on WR and, under heterogeneous values, need not dominate on J_B .
4. M5 will have negligible positive manipulation gain in the one-round validation and repeated independent setting, whereas M3 and M4 will remain vulnerable to bounded over-reporting.

The project can succeed even if the evidence rejects these hypotheses.

4.5 Anticipated limitations

The model omits job duration, multi-GPU jobs, placement, and preemption. Its findings apply to repeated unit-demand allocation, not to a production cluster. Synthetic values and utility-equivalent payments cannot establish external validity. Jain’s index captures one notion of equality, so the analysis also reports waiting and normalised-benefit metrics. Fixed attack policies do not characterise a multi-agent equilibrium, and the rollout attack is a finite-horizon unilateral search. The Vickrey result also depends on real quasi-linear transfers. Replacing payments with reusable credits would require a separate dynamic budget model and would remove the stated DSIC guarantee.

4.6 Success criteria and reproducibility

The project is *complete* when M1–M5 pass E0, E1–E4 run for every prespecified seed, paired intervals and raw results are produced, and one command can regenerate every main figure. E5–E6 and the rollout attack are stretch deliverables. The evidence must answer the research questions reliably, even when it contradicts the hypotheses.

5 Contribution

All members will review the formal model, experimental design, results, and final report. Each member has a primary area of responsibility.

Aayush Kuloor: simulator and reproducibility. Aayush will implement the valuation processes, histories, paired random streams, configuration loading, result storage, and end-to-end experiment runner. He will maintain the master seeds and reproducibility instructions.

Harsh Dhru: mechanisms and formal validation. Harsh will implement M1–M5, the allocation and payment interfaces, seeded tie-breaking, and the exhaustive E0 tests for mechanism invariants and the Vickrey benchmark.

Raj Modi: strategic policies and metrics. Raj will implement the bounded attacks and rollout diagnostic. He will also implement WR, J_A , J_B , waiting, starvation, PoF, PoS, and paired manipulation metrics, then validate them on hand-computed examples.

Kalp Shah: experiments and statistical analysis. Kalp will implement the parameter sweeps, paired bootstrap intervals, Pareto-dominance analysis, figures, and result tables. He will document failures, sensitivity checks, and positive findings.

Each owner will provide unit tests and a short methods description for their component. Another member will review that component’s code.

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